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Q.1 Write all four inequalities on w_1, w_2, and b required for perfect classification:

Ans = As per Truth Table, H(z) = 1 when $z \geq 0$, else 0. Here $z = w_1x_1 + w_2x_2 + b$. The conditions are:

Case 1: (x_1=0, x_2=0) requires y=0 $\Rightarrow b < 0$ (Eq. 1)

Case 2: (x_1=0, x_2=1) requires y=1 $\Rightarrow w_2 + b \geq 0$ (Eq. 2)

Case 3: (x_1=1, x_2=0) requires y=1 $\Rightarrow w_1 + b \geq 0$ (Eq. 3)

Case 4: (x_1=1, x_2=1) requires y=0 $\Rightarrow w_1 + w_2 + b < 0$ (Eq. 4)

Q.2 Prove, using those inequalities, that no choice of w_1, w_2, and b can classify every input correctly:

Ans = Proof: If we add (Eq. 2) and (Eq. 3) together:

$(w_2 + b) + (w_1 + b) \geq 0 + 0 \Rightarrow w_1 + w_2 + 2b \geq 0 \Rightarrow w_1 + w_2 + b \geq -b$

From (Eq. 1), we know $b < 0$, meaning $-b$ is positive (> 0).

So, $w_1 + w_2 + b > 0$. But (Eq. 4) strictly states $w_1 + w_2 + b < 0$.

This is a direct contradiction, hence a single perceptron cannot solve this.

Q.3 Construct a network with two hidden step neurons and one output step neuron. State all weights and biases.

Ans = A single perceptron fails, so we use an MLP with 2 hidden & 1 output neuron:

Hidden 1 (OR logic): w_{11}=1, w_{12}=1, b_1=-0.5 = $h_1 = H(x_1+x_2-0.5)$

Hidden 2 (AND logic): w_{21}=1, w_{22}=1, b_2=-1.5 = $h_2 = H(x_1+x_2-1.5)$

Output (h_1 AND NOT h_2): w_out_1=1, w_out_2=-2, b_out=-0.5 $\Rightarrow y = H(h_1-2h_2-0.5)$

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Q.4

Calculate the forward pass for all four samples and verify the truth table:

Ans =

Let's verify the forward pass for all 4 samples step-by-step:

Sample (0, 0): $h_1=H(-0.5)=0$, $h_2=H(-1.5)=0 \Rightarrow y=H(-0.5)=0$ [Verified]

Sample (0, 1): $h_1=H(0.5)=1$, $h_2=H(-0.5)=0 \Rightarrow y=H(0.5)=1$ [Verified]

Sample (1, 0): $h_1=H(0.5)=1$, $h_2=H(-0.5)=0 \Rightarrow y=H(0.5)=1$ [Verified]

Sample (1, 1): $h_1=H(1.5)=1$, $h_2=H(0.5)=1 \Rightarrow y=H(-1.5)=0$ [Verified]

The truth table is completely verified and matches the Desired alarm (y).

Q. 5

Explain why increasing training from 100 to 100,000 epochs cannot fix the original architecture, and distinguish representability from gradient-based trainability:

Ans =

Even if trained for 100,000 epochs, the single perceptron fails due to these reasons:

Representability (Can the model logically express the solution?):

A single perceptron only solves 'linearly separable' problems. XOR is not linearly separable. This means the original architecture fundamentally cannot represent the solution logically. Gradient-based Trainability (Can we find the optimal weights via training?):

Trainability finds the best weights via Gradient Descent. If the solution is missing logically, training forever will never find it.